METRIC

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DETAIL SPECIFICATION

ASPHALTIC COMPOUND, HIGH-MELTING HOTMELT (CAVITY LINING)

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high-melting hotmelt asphaltic compound for coating the inner walls of metal cases prior to filling with explosives (cavity lining).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-248 - Trinitrotoluene (TNT)

MIL-DTL-512 - Aluminum Powder, Flaked, Grained, and Atomized

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8030

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ASTM INTERNATIONAL

ASTM D5/D5M - Standard Test Method for Penetration of Bituminous Materials

ASTM D36/D36M - Standard Test Method for Softening Point of Bitumen (Ring-and-Ball Apparatus)

ASTM D92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester

ASTM D140/D140M - Standard Practice for Sampling Asphalt Materials

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.1.2 Materials. The compound shall be an asphaltic compound. The use of a polyamide wax as an additive to adjust viscosity of the molten compound shall be permissible. The materials used in the manufacture of the asphaltic compound shall produce a compound that is easily heat-liquefiable but does not flow at temperatures slightly below the melting point. The compound shall contain no volatile solvents.

3.1.3 Workmanship. The asphaltic compound shall be homogeneous and free from grit, lumps, contaminants, or other defects that could adversely affect its intended use. The asphalt compound shall be capable of storage at ambient temperature without release or separation of any component. The test procedure shall be in accordance with 4.4.2.

3.2 Product characteristics.

3.2.1 Chemical requirements. The asphaltic compound shall conform to the chemical requirements specified in table I when tested in accordance with 4.5.1.1 through 4.5.1.5.

TABLE I. Chemical requirements.

Requirement	Percent maximum
Manganese 1/	0.10
Lead 1/	0.04
Acidity, as sulfuric acid	0.01
Alkalinity, as sodium hydroxide	0.01
Insoluble matter in toluene	1.00
Ash	1.00

NOTE:

1/ Conformance tests (4.5.1.1 and 4.5.1.2) for manganese and lead are not required when the ash content is less than 0.004 percent.

3.2.2 Viscosity. The viscosity of the asphaltic compound shall be a maximum of 25 Zahn seconds at 232±2 °Celsius (C) when tested in accordance with 4.5.3.

3.2.3 Softening point stability. The softening point of the asphaltic compound shall be within the temperature range of 172 to 205 °C when tested in accordance with 4.5.4.

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3.2.4 Penetration. The asphaltic compound shall show the following penetration values when tested in accordance with 4.5.5:

a. 200 grams/60 seconds/0 °C 5 to 35 millimeters

b. 100 grams/5 seconds/25 °C 5 to 35 millimeters

c. 50 grams/5 seconds/46 °C 80 millimeters maximum

3.2.5 Flash point. The minimum flash point of the asphaltic compound shall be 300 °C when tested in accordance with 4.5.6.

3.3 Performance requirements.

3.3.1 Heat resistance. A dried film of the asphaltic compound on a metal panel shall not sag or flow when heated by immersion in a silicone oil bath maintained at 172±1 °C for 1 hour. The test procedure shall be in accordance with 4.5.7.

3.3.2 Acid resistance. A dried film of the asphaltic compound on a metal panel shall withstand the action of picric, sulfuric, nitric, and hydrochloric acids for 6 hours without disintegration, browning, or dulling. There shall be no etching of the metal underneath the compound. The test procedure shall be in accordance with 4.5.8.

3.3.3 Water resistance. A dried film of the asphaltic compound on a metal panel shall withstand immersion in distilled water for 18 hours without blistering, whitening, and not more than a slight dulling. The test procedure shall be in accordance with 4.5.9.

3.3.4 Flexibility. A dried film of the asphaltic compound on a metal panel shall show no chipping, scaling, or cracking when tested in accordance with 4.5.10.

3.4 Reactivity requirement. For first article inspection, or when specified (see 6.2), the reactivity for the asphaltic compound in contact with tritonal (80±3 parts TNT in accordance with MIL-DTL-248, and 20±3 parts aluminum in accordance with MIL-DTL-512) explosive in the vacuum stability test of 4.5.11 shall not exceed 3.0 milliliters (mL) of gas (over and above that generated by the controls).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (table II lists the applicable tests required for each particular level of inspection):

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

TABLE II. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Workmanship	X	X	3.1.3	4.4.2
Chemical requirements	X	X	3.2.1	4.5.1
Viscosity	X	X	3.2.2	4.5.3
Softening point stability	X	X	3.2.3	4.5.4
Penetration	X	X	3.2.4	4.5.5
Flash point	X	X	3.2.5	4.5.6
Heat resistance	X	X	3.3.1	4.5.7
Acid resistance	X	X	3.3.2	4.5.8
Water resistance	X	X	3.3.3	4.5.9
Flexibility	X	X	3.3.4	4.5.10
Reactivity	X	---	3.4	4.5.11

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4.2 First article inspection. When required (see 3.1.1), the first article inspection shall include the tests required specified in table II. Failure of the first article sample to pass this inspection shall be cause for rejection of the first article sample.

4.2.1 First article sample. The first article sample shall consist of 4.54 kilograms (10 pounds) of the asphaltic compound manufactured and taken from one batch of the material. If the sample is provided to the Government as specified (see 6.2) or to a laboratory acceptable to NAVSEA for inspection, the first article sample container shall be marked with the following information:

a. Asphaltic compound, high-melting, hotmelt (cavity lining).

b. Formulation and identity of additives, if any, to the asphaltic material.

c. Name and address of the manufacturer.

d. Location and identity of the plant that produced the samples.

e. Manufacturer's identification.

f. Date of manufacture.

g. Submitted by (name) (date) for first article tests in accordance with the requirements of MIL-DTL-82604.

4.3 Conformance inspection. Conformance inspection shall include the tests specified in table II, as well as an examination of filled containers (see 4.3.3) for conformance to packaging, packing, and marking requirements. The reactivity requirement of 3.4 may be a requirement for conformance inspection if it is determined the material submitted under this specification is suspect (see 6.2).

4.3.1 Inspection lot. An inspection lot shall consist of the same type asphaltic compound produced by one manufacturer, at one plant, from the same materials and formulation, and under the same manufacturing conditions.

4.3.2 Sample for tests. A 4.54-kilogram (10-pound) sample of asphaltic compound shall be taken at random from each lot in accordance with ASTM D140/D140M. The sample material shall be chosen at random from containers of "solidified" compound. A sample for the tests of 4.5.3 through 4.5.10 shall be preconditioned for testing in accordance with 4.5.2. The sample for tests of 4.5.1 need not be preconditioned. The lot shall be rejected if it fails to meet all the requirements of this specification, except as noted in 4.3.

4.3.2.1 Sampling. Each container shall be labeled with complete information on the lot and batch number, date of sampling, contract number, and the applicable revision of MIL-DTL-82604.

4.3.3 Sample for examination of filled containers. A random sample of filled containers shall be selected from each lot of compound in accordance with MIL-STD-1916, verification level II.

4.4 Inspection procedure.

4.4.1 General. The values specified in section 3 shall apply to the results obtained from one determination for each particular test.

4.4.2 Visual examination. Conformance of the asphaltic compound to the requirement for workmanship (see 3.1.3) and any other requirements not covered by specific tests shall be determined by appropriate visual examination using daylight illumination.

4.4.3 For examination of preparation for delivery. Using the sample of filled containers selected in 4.3.3, the sample shall be in accordance with MIL-STD-1916, verification level II. The sample unit shall be one filled and closed shipping container. Sample containers shall be examined for compliance with all requirements of this specification in regard to contents, closure, damaged or leaking container, improper container, and marking.

4.5 Test methods and procedures.

4.5.1 Chemical requirements. The chemical requirements of 3.2.1 shall be determined as follows (see 4.5.1.1 through 4.5.1.5).

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4.5.1.1 Manganese. Weigh a 10-gram sample of the compound, dissolve in 10 mL of concentrated nitric acid, and filter if necessary. Dilute to 100 mL with distilled water and transfer a 20-mL aliquot of the solution to a 250-mL beaker. Add 1 gram of lead peroxide and boil. Note if manganese is present by the development of a red coloration in the solution. If the presence of manganese is indicated, filter through asbestos and titrate the clean filtrate with N/10 oxalic acid until the red coloration disappears. Calculate the percentage of manganese in the sample as follows:

$$\text{Percentage of manganese} = \frac{1.099VN}{W}$$

Where: V = mL of N/10 oxalic acid used
N = normality of oxalic acid
W = weight of sample

4.5.1.2 Lead. Transfer the remaining 80-mL portion of the solution obtained as described in 4.5.1.1 to a beaker and evaporate to 5 mL. Make the solution slightly alkaline with ammonium hydroxide. Acidify with acetic acid and add an excess. Heat to the boiling point and add 10 to 15 mL of a 10-percent solution of sodium or potassium dichromate. Boil the solution and allow to stand for 16 hours. Note if lead is present as indicated by a yellow precipitate. If lead is present, filter the solution through a tared Gooch crucible (or equivalent) and wash the precipitate with water and alcohol. Dry the crucible and contents at approximately 110 °C (230 °F) for 1 hour, cool, and weigh. Calculate the increase in weight of the crucible as lead in the sample as follows:

$$\text{Percent lead} = \frac{64.1A}{W}$$

Where: A = weight of residue, gram
W = weight of original sample, gram

4.5.1.3 Acidity or alkalinity. Dissolve a 5-gram sample of the compound in 10 mL of toluene, neutral to phenolphthalein. Shake the solution with 100 mL of distilled water in a 250 mL separatory funnel for at least 5 minutes. Filter and titrate water phase with N/100 sulfuric acid or N/100 sodium hydroxide as necessary using phenolphthalein as indicator. Run a blank on an equal volume of the water used.

4.5.1.4 Insoluble matter in toluene.

4.5.1.4.1 Apparatus. Use an extraction apparatus, compatible for use with an electric hotplate or steam bath, that consists of a heat resistant glass assay flask, a block tin condensing coil with a nickel, copper, or tin-plated metal cap that fits securely around the rim of the flask, and a glass siphon cup that is suspended under the condenser.

4.5.1.4.2 Method. Transfer a sample of approximately 6 grams, weighed accurately to 0.1 mg, into a tared stainless steel 200-mesh thimble. Fabricate the thimble from a 15-centimeter diameter circle of wire mesh by using the lower portion of a test tube as a form for folding the thimble. This form should be of a diameter that will fit snugly into the glass siphon tube. Place the thimble containing the sample into the siphon cup. Do not allow the thimble to extend beyond the top of the cup. Suspend the cup under the condenser by a fine wire bail. Add approximately 60 mL of toluene to the sample in the cups and allow the excess solvent to siphon over into the flask. Put the condenser in place and circulate cold water through it. Heat the flask on a steam bath or an electric hotplate. Allow the solvent to drip from the top of the condenser at a slow to medium rate. Discontinue the extraction when the solvent drops colorless from the siphon cup. Wash down the steel mesh with a fine jet of solvent from a wash bottle. If the washings show any color, put the sample back into the siphon cup and continue the extraction until the solvent again becomes colorless. Remove the steel mesh thimble containing the residue, dry for 1 hour at approximately 100 °C, cool, and weigh. Calculate the percent insoluble in the solvent as follows:

$$\text{Percent insoluble in toluene} = \frac{A - B}{W} \times 100$$

Where: A = weight of steel mesh thimble plus residue after extraction, gram
B = weight of steel thimble, gram
W = weight of sample, gram

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4.5.1.5 Ash. Transfer a weighed portion of approximately 1 gram of the sample to a tared porcelain or platinum crucible and place in a cold muffle furnace having good air circulation and capable of temperature regulation between 700 and 750 °C. Gradually raise the temperature of the furnace to a red heat. Finish the ignition to constant weight at a temperature between 700 and 750 °C. Cool in a desiccator and weigh. Calculate the weight of residue as percent ash in the sample.

4.5.2 Test sample preparation. The material for all of the tests required by 3.2.2 through 3.3.4 shall be prepared at the same time and from the same melt of the test sample selected in accordance with 4.3.2 as follows.

4.5.2.1 Apparatus.

a. Sample container - Stainless steel, 1-liter beaker approximately 10.48 centimeters (4⅛ inches) inside diameter by 14.60 centimeters (5¾ inches) high, straight sides, no pouring lip.

b. Oven - Electric, mechanical convection type, temperature range: ambient to 250 °C.

c. Stirring rod - 0.79 centimeter ($\frac{5}{16}$-inch) diameter, 30.48 centimeters (12 inches) long, polytetrafluoroethylene (PTFE) with a steel core.

4.5.2.2 Preconditioning. Transfer a 600- to 650-gram sample of the asphaltic compound selected in 4.2 to a 1-liter stainless steel beaker. Heat the sample in an oven, mechanical convection type, for 3¾ hours. Set the temperature of the oven to 232±2 °C. When the oven is at the desired temperature, place the sample in the center of the oven with good air circulation on all sides. Hand-stir the sample with the PTFE stirring rod every 30 minutes to prevent lumping. At the end of the heating period, remove the sample from the oven for the Zahn viscosity test of 4.5.3. When this test is completed, return the sample and its container to the same oven and heat for an additional 30 minutes. Do not allow the sample to remain out of the oven for viscosity testing for longer than 2 minutes. At the end of the second heating period, pour the samples and coat steel panels for the tests of 4.5.4 through 4.5.10.

4.5.3 Viscosity. The viscosity of the asphaltic compound shall be determined with a Zahn viscosimeter, No. 5 cup, at the end of the 3¾ hours of heating at 232±2 °C. The viscosimeter shall be preheated in the same oven and at the same time as the sample is being prepared. The sample beaker containing the molten compound and the viscosimeter shall be removed from the oven. The viscosimeter shall be held in a vertical position and immediately dipped into the sample completely filling the cup. The cup shall be lifted 1.27 centimeters (½ inch) above the top surface of the beaker and the asphaltic compound shall be allowed to drain into the beaker. With a stopwatch, the time shall be recorded in seconds between when the cup breaks the surface of the sample and when the steady flow of the liquid from the orifice suddenly stops.

4.5.4 Softening point stability. The softening point shall be determined on two samples run in the same bath at the same time in accordance with ASTM D36/D36M using the double ring-holding device of ASTM D36/D36M, except Dow Corning Silicone Fluid PMX-200 of 350 centistokes viscosity, or equivalent (see 6.5), shall be used in lieu of water for the bath fluid. The thermometer shall be a total immersion, range -10 to 250 °C, 0.5-degree divisions, 40.64 centimeters (16 inches) in length. Two shouldered rings shall be employed and the samples shall be heated at a rate of 5 °C per minute ±1 °C for the last 30 degrees before the softening point is reached. The top cover of the ring-holding device shall be drilled to accommodate a small glass, motor driven stirrer with a 15.24-centimeter (6-inch) shaft. The shouldered rings shall be heated to 232±2 °C before the molten compound is poured. To facilitate the release of the filled rings from the bottom plate of the mold, the plate shall be wiped with silicone oil, leaving a thin film before pouring the sample. After the samples have solidified, the mold shall be placed on a hotplate. When the mold becomes hot, but can still be handled with the bare hand, the mold shall be removed from the hotplate and the rings shall be slid from the bare plate to a piece of aluminum foil. The average of the values shall be taken to determine the softening point.

4.5.5 Penetration. The penetration values shall be determined in accordance with ASTM D5/D5M. Ointment cups shall be preheated before pouring samples into them.

4.5.6 Flash point. The flash point shall be determined in accordance with ASTM D92.

4.5.7 Heat resistance. Preparation of the test panel and method for determining heat resistance shall be as follows.

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4.5.7.1 Preparation of test panel. Use a solvent-cleaned, mechanically-polished, plain carbon steel panel, 15.24 by 10.16 centimeters (6 by 4 inches), by approximately 0.32 centimeter (⅛ inch) thick. Preheat the panel for 5 minutes, prior to dipping in the preconditioning oven (see 4.5.2.2). Coat the panel by dipping it into the molten compound, which shall be maintained at 232±2 °C. Leave the top portion of the panel uncoated. Ensure the asphalt thickness is approximately 0.16 centimeter (1/16 inch) on each side of the panel. Suspend the panel with the uncoated portion up and allow to cool to room temperature. Trim the excess compound from the bottom edge of the panel with a sharp knife.

4.5.7.2 Method. Suspend a panel, prepared as described in 4.5.7.1, in a bath Dow Corning Silicone Fluid PMX-200 of 350 centistokes viscosity, or equivalent (see 6.5), at 172±1 °C for 1 hour so that the uncoated end of the panel is uppermost and one-half of coated portion is submerged. Remove the panel and cool by immersion in cold water. Examine the film for sag or flow. A slight surface displacement at the silicone line is permissible.

4.5.8 Acid resistance. A panel, prepared as described in 4.5.7.1, shall be placed on a flat horizontal surface. Drops of the following solutions shall be placed at different places on the coating:

a. Saturated picric acid

b. Sulfuric acid, specific gravity 1.30

c. Nitric acid, specific gravity 1.22

d. Hydrochloric acid, specific gravity 1.09

The acids shall be allowed to remain in contact with the coating for 6 hours while the temperature is maintained at approximately 21 °C. The drops shall be covered with small watch glasses in order to prevent evaporation. The acids shall be removed by washing with cold water and the film shall be dried at room temperature for 1 hour. The areas previously in contact with the acids shall be examined and noted if the compound is resistant to each of these acids as indicated by absence of disintegration, browning, or dulling of the coating. A slight bloom around the area of a spot exposed to acid shall not be considered an indication of failure to resist the acid. The coating shall be removed with benzene and the metal shall be examined. It shall be noted if the coating has resisted the action of each acid as indicated by the absence of any corrosion of the metal.

4.5.9 Water resistance. A panel prepared as described in 4.5.7.1 shall be suspended in a beaker of cold distilled water for 18 hours so that the uncoated end of the panel is uppermost and one-half of the coated portion is submerged. The panel shall be examined for blistering, whitening, softening, or other visible defects.

4.5.10 Flexibility. A strip of No. 30-gage hot-rolled steel sheet, 5.08 by 20.32 centimeters (2 by 8 inches), shall be cleaned by washing with benzol and thoroughly dried. A preheated strip shall be coated on one side to a thickness of 0.16 centimeter (1/16 inch) by flowing on the molten compound and then cooled at room temperature for a period of 6 hours. With the coating on the outside, the test panel shall be bent rapidly 180 degrees over a mandrel 2.54 centimeters (1 inch) in diameter. No macroscopic cracking of any kind shall be visible after the bending test.

4.5.11 Reactivity. The reactivity of the asphaltic compound in contact with tritonal explosives shall be determined in accordance with the vacuum stability test procedures of 4.5.11.1 through 4.5.11.4.

4.5.11.1 Preparation of samples. Heat the compound until pourable, then deposit on glass plates an amount of the compound to provide 2.5 (N+2) grams of dried film (where N equals the number of explosives involved). Air-dry the films under ambient conditions for 48 hours. After drying, remove and cut into approximately 0.64-centimeter (¼-inch) squares. Conduct this operation in a room that is free from acid, alkaline, or other contaminating fumes. Reduce the explosive materials to 12 mesh or finer by grinding or rasping. Keep both the compound film and the ground explosives dry by storing in a desiccator over a desiccating agent until ready for testing.

4.5.11.2 Calibration of apparatus. Calibrate the necessary number of vacuum stability test assemblies in accordance with MIL-STD-286, method 403.1.3.

4.5.11.3 Testing procedure. Use 2N + 2 (where N equals the number of explosives used) tubes similar to the heating tube portion of the apparatus in accordance with MIL-STD-286, method 403.1.3. For controls, add 2.5 grams of the dried compound to each of two tubes and 2.5 grams of each explosive to additional individual tubes. Place uniform mixtures of 2.5/2.5 grams of the asphaltic compound and each of the explosives specified in the test in single separate tubes. Obtain uniform distribution or mixing by carefully layering alternate portions of the dried compound with the explosive. After the addition of each portion, lightly tamp the mixture with a glass or

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metal rod to ensure intimate contact of the materials. After all the samples have been prepared, connect the respective heating tubes with the capillary tubes as prescribed and conduct the vacuum stability test in accordance with MIL-STD-286, method 403.1.3, at 100±0.5 °C for 48 hours. Make all readings at room temperature with the samples removed from the bath. Correct the readings of both the controls and the test samples to standard conditions of temperature and pressure, and check for compliance with 3.4.

4.5.11.4 Calculation of reactivity. Calculate the reactivity of each of the explosive materials with the coating compound as follows:

$$\text{Reactivity in mL of gas} = X - (Y + Z)$$

Where: X = mL of gas produced by the mixture of explosive material and asphaltic compound
Y = mL of gas produced by the explosive material alone
Z = mL of gas produced by the asphaltic compound alone

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The high-melting, hotmelt asphaltic compound covered by this specification is intended for use as a protective lining for fabricated metal cases in applications where charges of high explosives are poured into the case and require thermal insulation from the elevated temperature environment of a fuel fire. As this compound was developed for use in military ordnance, there is no commercial application. Existing stock of the high-melting, hotmelt asphaltic compound is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity of asphaltic compound ordered.
c. When first article inspection is required (see 3.1.1 and 6.3).
d. Whether reactivity is required (see 3.4).
e. Testing activity designated to perform first article tests.
f. Type and degree of contractor quality assurance program required.
g. Packaging requirements (see 5.1 and 6.4).
h. Marking requirements (see 6.4.2).
i. Activities requiring copies of completed Safety Data Sheets (SDS) (see 6.4.2.2).

6.3 First article. When first article inspection is required, the contracting officer will provide specific guidance to offerors whether the sample(s) will be a preproduction sample, a first article sample, a first production sample, a sample selected from the first production batches, or a standard production sample(s) from the contractor's current inventory. The contracting officer will also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first article samples. Invitations for bids will provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation. Generally, a first article inspection, or part of a first article inspection, will be waived only if the following requirements have been met:

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a. There has been no interruption in the manufacturer's production of asphaltic compound exceeding 12 months at the time of contract award.

b. The manufacturer has made no changes to their process, facilities, procedures, or ingredient manufacturers since last qualified.

6.4 Preparation for delivery.

6.4.1 Packaging and packing. The material should be packed in type I, grade C fiber drums conforming to PPP-D-723. The inside of the container should be treated with a compatible release agent that will not contaminate the contents and will promote ready removal of the contents. Each drum should be furnished with a top heading. Drums should be palletized in accordance with MIL-STD-147.

6.4.2 Marking. Unless otherwise specified (see 6.2), marking should be in accordance with MIL-STD-290, as applicable.

6.4.2.1 Hazardous material identification. In addition to the marking specified in MIL-STD-290, the fiber drums should be marked with hazardous material identification symbols in accordance with FED-STD-313.

6.4.2.2 SDSs. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.5 Silicone fluid of 350 centistokes viscosity. Silicone fluid of 350 centistokes viscosity is available from Dow Corning (CAGE Code 5D028), www.dow.com, 989-636-1000, 2211 H.H. Dow Way Midland, MI 48674 as XIAMETER™ PMX-200 Silicone Fluid 350cSt, or may be available from equivalent suppliers.

6.6 Previous sources of supply. The following listed manufacturers were previously approved sources and may no longer produce a product that meets the requirements of this specification.

Government Designation	Manufacturer's Designation	Test Reference	Manufacturer's Name and Address
MIL-C-82604	"Pioneer 590L"	NEDE Dept. (Code 502) Jan 1973	Chemtura 4240 White Oak Ave. Hammond, IN 46320 Plant: Lawrenceville, IL www.chemtura.com/ 800.243.3024
MIL-C-82604	"Pioneer 590P"	NEDE Dept. (Code 502) Jan 1973	Chemtura 1000 Convery Bldg. Perth Amboy, NJ 06861 Plant: Perth Amboy, NJ www.chemtura.com/ 800.243.3024

6.7 Subject term (key word) listing.

Asbestos
High explosives
Thermal insulation

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 8030-2024-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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